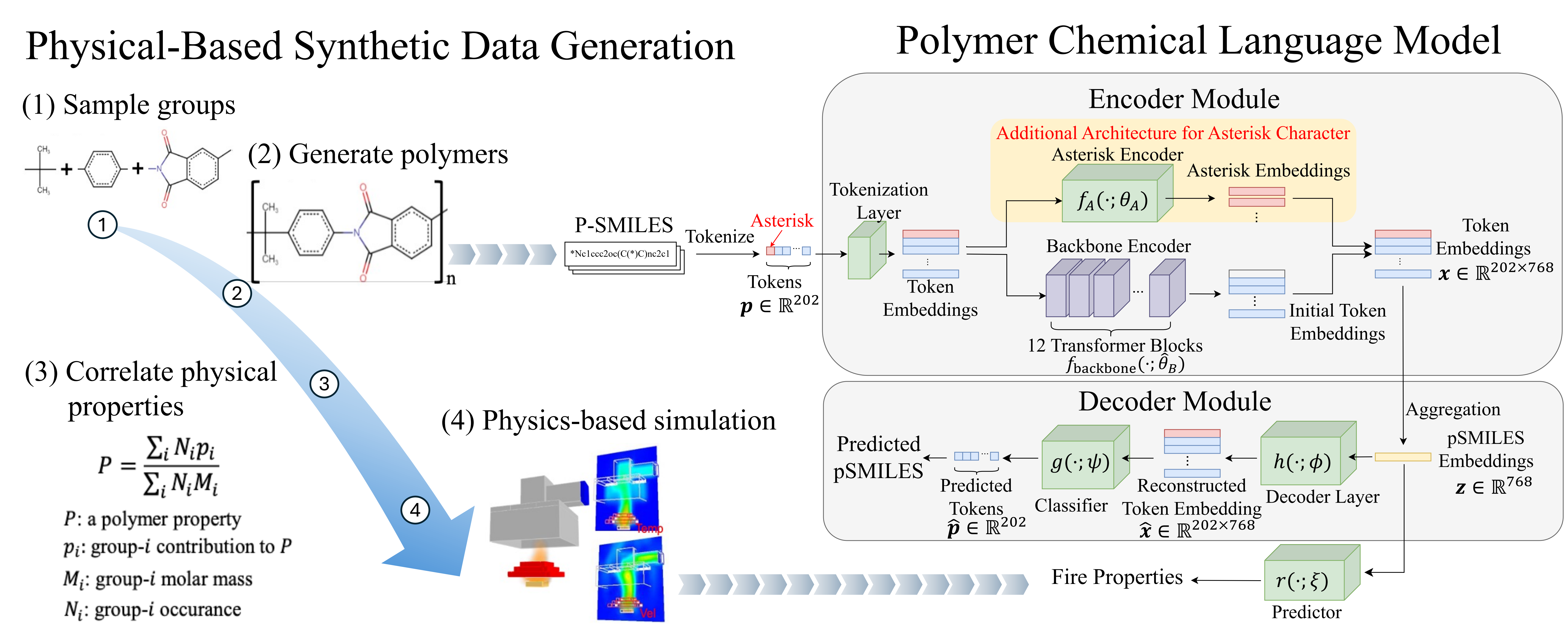
Physical-Based Synthetic Data Generation
Polymer Chemical Language Model
(1) Sample groups
(2) Generate polymers
n
P-SMILES
*Nc1ccc2oc(C(*)C)nc2c1
Tokenize
Asterisk
Tokens
$\boldsymbol{p} \in \mathbb{R}^{202}$
(3) Correlate physical properties
$P = \frac{\sum_i N_i p_i}{\sum_i N_i M_i}$
P: a polymer property
p_i: group-i contribution to P
M_i: group-i molar mass
N_i: group-i occurance
(4) Physics-based simulation
Temp
Vel
1
2
3
4
Encoder Module
Additional Architecture for Asterisk Character
Asterisk Encoder
$f_A(\cdot;\theta_A)$
Asterisk Embeddings
Tokenization Layer
Token Embeddings
Backbone Encoder
12 Transformer Blocks
$f_{\text{backbone}}(\cdot;\hat{\theta}_B)$
Initial Token Embeddings
Token Embeddings
$\boldsymbol{x} \in \mathbb{R}^{202\times768}$
Aggregation
Decoder Module
Predicted pSMILES
Predicted Tokens
$\hat{\boldsymbol{p}} \in \mathbb{R}^{202}$
$g(\cdot;\psi)$
Classifier
Reconstructed Token Embedding
$\hat{\boldsymbol{x}} \in \mathbb{R}^{202\times768}$
$h(\cdot;\phi)$
Decoder Layer
pSMILES Embeddings
$\boldsymbol{z} \in \mathbb{R}^{768}$
Fire Properties
$r(\cdot;\xi)$
Predictor